Lecture 5 (Affine and Convex Functions)

January 27, 2026

We will talk about two special classes of functions: Affine and Convex functions

Affine Functions

Affine functions are functions $f : \mathbb{R}^n \to \mathbb{R}^m$ of the form

$$f(\vec{x}) = A\vec{x} + \vec{b}$$

for some $A \in \mathbb{R}^{m \times n}$ and $\vec{b} \in \mathbb{R}^m$. These are a generalization of linear transformations. Geometrically, these can be thought of as linear transformations with translation allowed. These are useful to use, as many of our constraints in LPs can be thought of as affine functions.

Properties

Let's prove some properties of Affine Functions. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be affine. Then:

1. $f\left(\vec{0}\right) = \vec{b}$
2. $f(\vec{x}) - f\left(\vec{0}\right) = g(\vec{x})$ is linear
3. Linear combinations of affine functions are affine
4. composition of affine functions are affine.

Proofs

1)

$$f\left(\vec{0}\right) = A\vec{0} + \vec{b} = \vec{0} + \vec{b} = \vec{b}$$

2)

$$f(\vec{x}) - f\left(\vec{0}\right) = \left(A\vec{x} + \vec{b}\right) - \left(\vec{b}\right) = A\vec{x}$$

3)

Let $f_i : \mathbb{R}^n \to \mathbb{R}^m$ be affine for $1 \leq i \leq n$. Let $c_i \in \mathbb{R}$ for $1 \leq i \leq n$. Let

$$f_i(\vec{x}) = A_i\vec{x} + \vec{b_i}$$

Define

$$f(\vec{x}) = \sum_{i=1}^{n} c_i f_i(\vec{x})$$

to be a linear combination of affine functions. Then

$$f(\vec{x}) = \sum_{i=1}^{n} c_i \left(A_i \vec{x} + \vec{b_i} \right) = \sum_{i=1}^{n} c_i A_i \vec{x} + c_i \vec{b_i} = \sum_{i=1}^{n} c_i A_i \vec{x} + \sum_{i=1}^{n} c_i \vec{b_i}$$

$$= \left(\sum_{i=1}^{n} c_i A_i \right) \vec{x} + \left(\sum_{i=1}^{n} c_i \vec{b_i} \right) = A\vec{x} + \vec{b}$$

Where $A = \sum_{i=1}^{n} c_i A_i$ and $\vec{b} = \sum_{i=1}^{n} c_i \vec{b_i}$.

4)

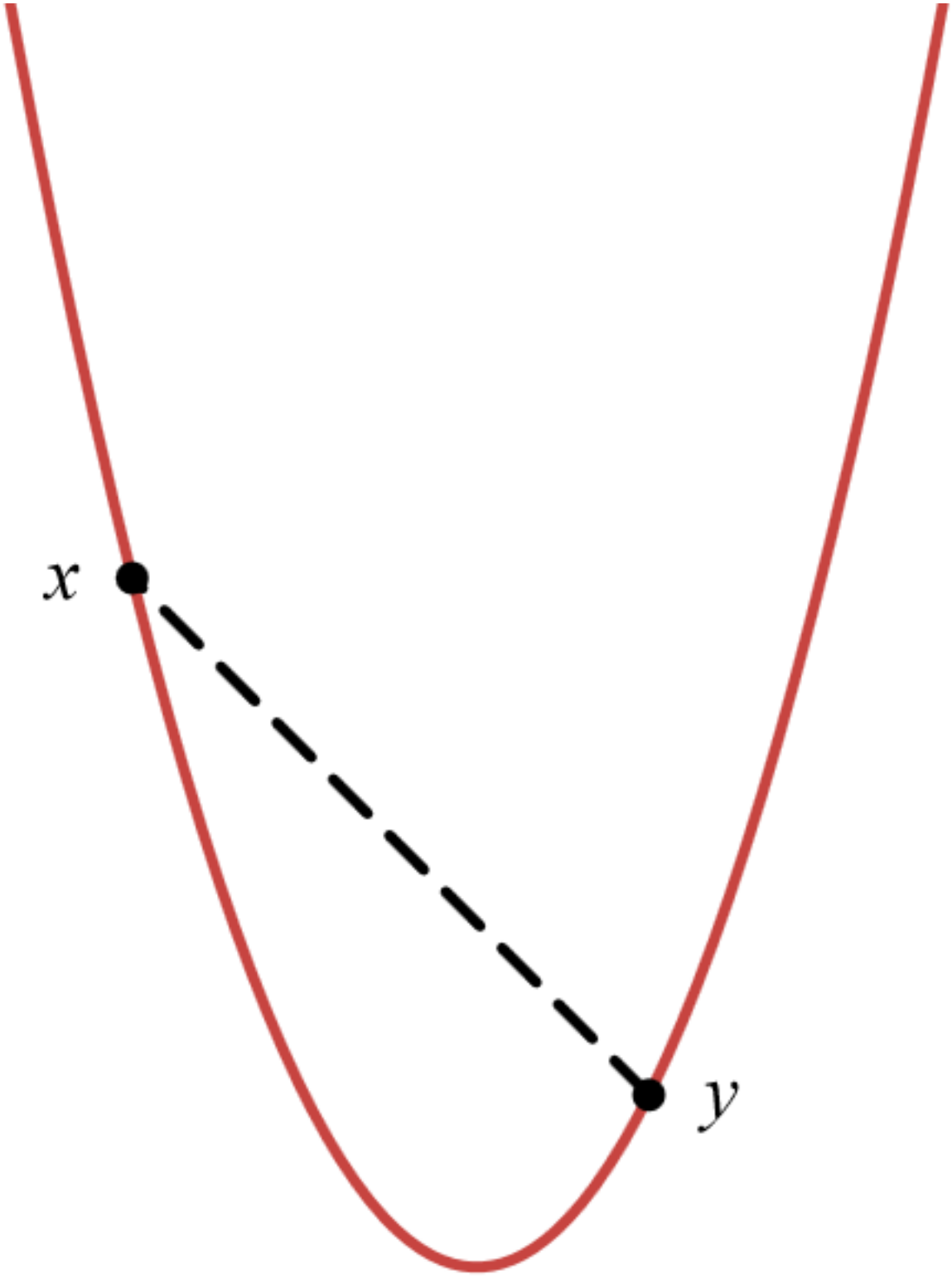
Left as homework exercise.

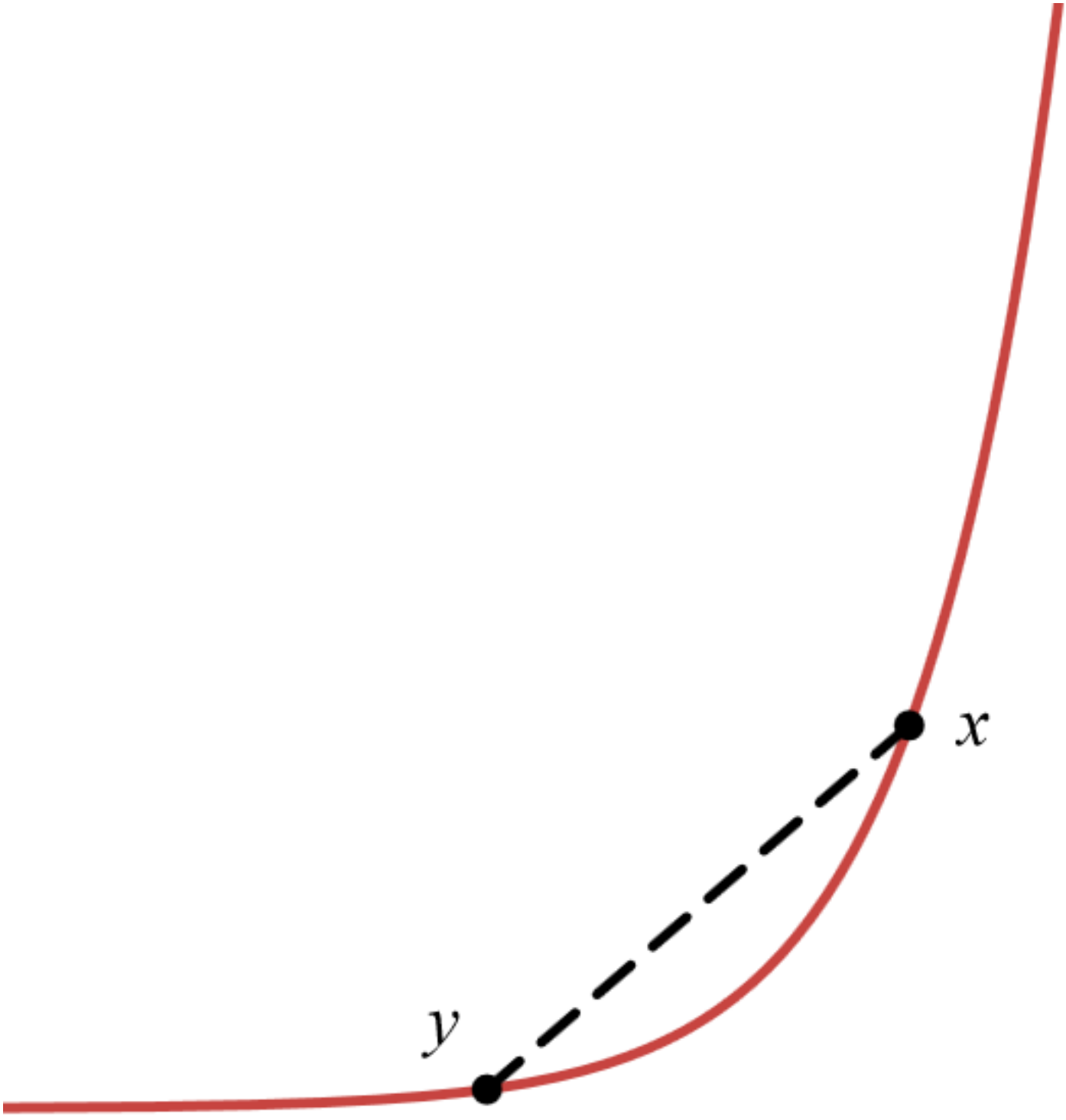
Convex Functions

Convex functions have a trickier definition, but it turns out the convexity yields many useful properties for optimization. A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be **convex** if each $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$ obey the following inequality:

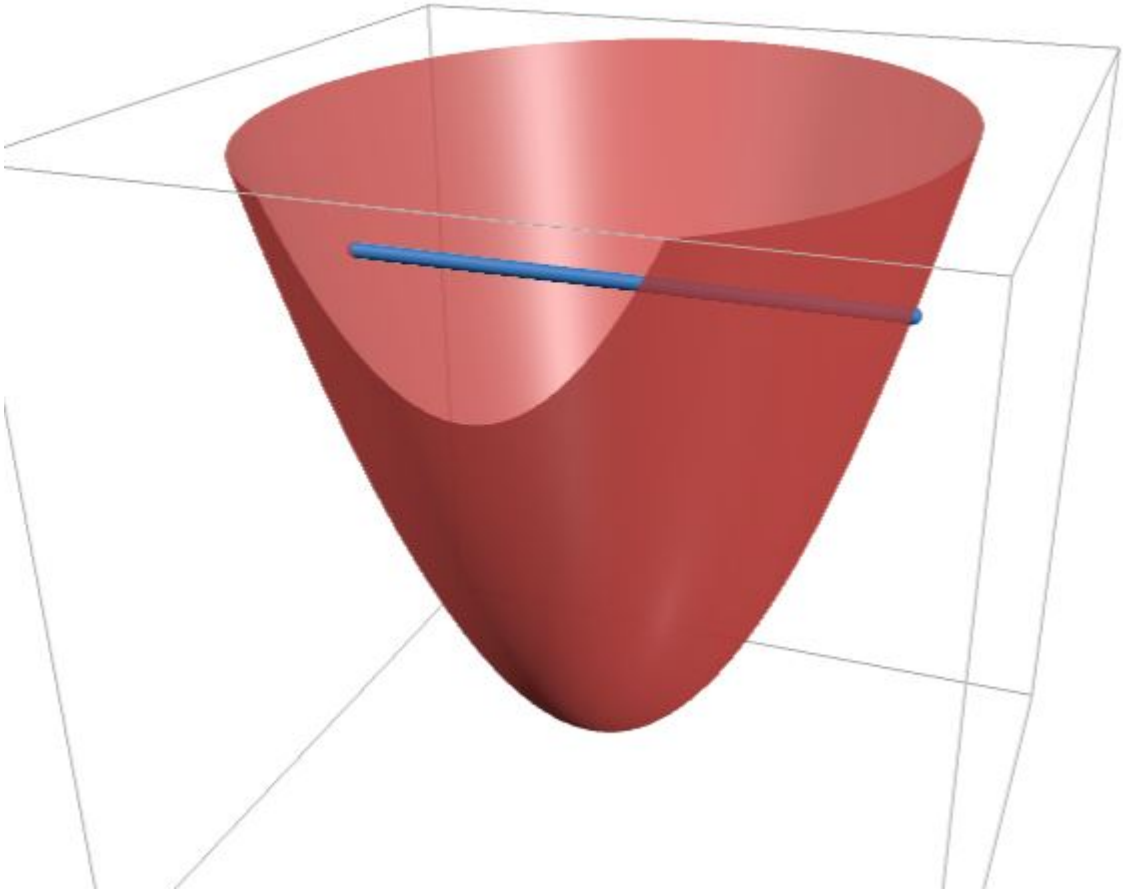
$$f(\lambda \vec{x} + (1 - \lambda)\vec{y}) \leq \lambda f(\vec{x}) + (1 - \lambda) f(\vec{y})$$

The intuition here is that for any two points in the domain of f, the line segment joining their outputs must lie on or above the function's image. In 2d, a couple of examples are shown below:


x
y


x
y

An example in 3D is also shown below (Note it is a little harder to see since Desmos 3D isn't as powerful as Desmos 2D currently)



Examples

Let's give a few examples of convex functions.

Ex 1)

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is affine. Then f is convex.

Proof

Suppose $f(\vec{x}) = A\vec{x} + b$ and let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and let $\lambda \in [0, 1]$. Then

$$f(\lambda\vec{x} + (1-\lambda)\vec{y}) = A(\lambda\vec{x} + (1-\lambda)\vec{y}) + b = \lambda A\vec{x} + (1-\lambda)A\vec{y} + b$$

$$= \lambda A\vec{x} + (1-\lambda)A\vec{y} + \lambda b + (1-\lambda)b = \lambda(A\vec{x} + b) + (1-\lambda)(A\vec{y} + b) = \lambda f(\vec{x}) + (1-\lambda)f(\vec{y})$$

Ex 2)

Let $f(\vec{x}) = \|\vec{x}\|^2$. Then f is convex.

Proof

First observe the following:

$$0 \leq (\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y}) = \|\vec{x}\|^2 - 2\vec{x} \cdot \vec{y} + \|\vec{y}\|^2$$

Thus

$$2\vec{x} \cdot \vec{y} \leq \|\vec{x}\|^2 + \|\vec{y}\|^2$$

Call this inequality (*). Now consider:

$$f(\lambda\vec{x}+(1-\lambda)\vec{y}) = (\lambda\vec{x}+(1-\lambda)\vec{y})\cdot(\lambda\vec{x}+(1-\lambda)\vec{y}) = \lambda^2\|\vec{x}\|^2 + \lambda(1-\lambda)\vec{x}\cdot\vec{y} + (1-\lambda)^2\|\vec{y}\|^2$$

by (*), we see

$$\leq \lambda^2\|\vec{x}\|^2 + \lambda(1-\lambda)\left(\|\vec{x}\|^2 + \|\vec{y}\|^2\right) + (1-\lambda)^2\|\vec{y}\|^2 = \lambda^2\|\vec{x}\|^2 + \lambda(1-\lambda)\|\vec{x}\|^2 + \lambda(1-\lambda)\|\vec{y}\|^2 + (1-\lambda)^2\|\vec{y}\|^2$$

$$= \lambda\|\vec{x}\|^2(\lambda+(1-\lambda)) + (1-\lambda)\|\vec{y}\|^2(\lambda+(1-\lambda)) = \lambda\|\vec{x}\|^2 + (1-\lambda)\|\vec{y}\|^2$$

So $f(\vec{x})$ is convex. A similar argument can also be used to show that $f(\vec{x}) = \langle\vec{x},\vec{x}\rangle$ is convex for any inner product, $\langle\cdot,\cdot\rangle$. This will include all of the LP norms, positive integer powers, absolute values, and the like.

Ex 3)

A matrix $A \in \mathbb{R}^{n\times n}$ is said to be **positive semi-definite** if for all $\vec{x} \in \mathbb{R}^n$ we have

$$0 \leq \vec{x}^T A\vec{x}$$

It turns out this is equivalent to matrix A containing all positive eigenvalues. It turns out that for any matrix A the product:

$$\vec{x}^T A\vec{x} = \vec{x}^T\left(\frac{A+A^T}{2}\right)\vec{x}$$

(See if you can prove this using index notation!). We call $\vec{x}^T A\vec{x}$ the **quadratic form**. As $\left(\frac{A+A^T}{2}\right)$ is symmetric, we will safely assume A is symmetric (when working with quadratic forms). It turns out

$$f(\vec{x}) = \vec{x}^T A\vec{x}$$

is convex if and only if A is positive definite.

Proof

First, suppose A is positive definite (and symmetric). Then

$$0 \leq (\vec{x}-\vec{y})^T A(\vec{x}-\vec{y}) = \vec{x}^T A\vec{x} - \vec{y}^T A\vec{x} - \vec{x}^T A\vec{y} + \vec{y}^T A\vec{y}$$

but as A is symmetric, we have $A_{ij} = A_{ji}$. In index notation, notice that

$$\vec{y}^T A\vec{x} = y_i A_{ij} x_j = y_i A_{ji} x_j = x_j A_{ji} y_i = \vec{x}^T A\vec{y}$$

so we see that for any quadratic form, $\vec{x}^T A\vec{y} = \vec{y}^T A\vec{x}$ and thus:

$$0 \leq \vec{x}^T A\vec{x} - 2\vec{x}^T A\vec{y} + \vec{y}^T A\vec{y}$$

$$2\vec{x}^T A\vec{y} \leq \vec{x}^T A\vec{x} + \vec{y}^T A\vec{y}$$

Now, consider

$$f(\lambda\vec{x}+(1-\lambda)\vec{y}) = (\lambda\vec{x}+(1-\lambda)\vec{y})^T A(\lambda\vec{x}+(1-\lambda)\vec{y})$$

$$= \lambda^2\vec{x}^T A\vec{x} + 2\lambda(1-\lambda)\vec{x}^T A\vec{y} + (1-\lambda)^2\vec{y}^T A\vec{y} \leq \lambda^2\vec{x}^T A\vec{x} + \lambda(1-\lambda)\left(\vec{x}^T A\vec{x} + \vec{y}^T A\vec{y}\right) + (1-\lambda)^2\vec{y}^T A\vec{y}$$

$$= \lambda\vec{x}^T A\vec{x}(\lambda+(1-\lambda)) + (1-\lambda)\vec{y}^T A\vec{y}(\lambda+(1-\lambda)) = \lambda\vec{x}^T A\vec{x} + (1-\lambda)\vec{y}^T A\vec{y} = \lambda f(\vec{x}) + (1-\lambda)f(\vec{y})$$

Thus f is convex. Now, suppose A is not positive semi-definite. This means there is a $\vec{x}$ such that $f(\vec{x}) = \vec{x}^T A\vec{x} < 0$. This quadratic form will form a polynomial if written out. As such, we know f is continuous and there must exist a vector $\vec{y}$ near $\vec{0}$ such that

$$f(\vec{x}-\vec{y}) < 0$$

As such, we see

$$0 > (\vec{x}-\vec{y})^T A(\vec{x}-\vec{y}) = \vec{x}^T A\vec{x} - 2\vec{x}^T A\vec{y} + \vec{y}^T A\vec{y}$$

$$2\vec{x}^T A\vec{y} > \vec{x}^T A\vec{x} + \vec{y}^T A\vec{y}$$

So

$$f(\lambda\vec{x} + (1-\lambda)\vec{y}) = (\lambda\vec{x} + (1-\lambda)\vec{y})^T A(\lambda\vec{x} + (1-\lambda)\vec{y})$$

$$\lambda^2\vec{x}^T A\vec{x} + 2\lambda(1-\lambda)\vec{x}^T A\vec{y} + (1-\lambda)^2\vec{y}^T A\vec{y} > \lambda^2\vec{x}^T A\vec{x} + \lambda(1-\lambda)\left(\vec{x}^T A\vec{x} + \vec{y}^T A\vec{y}\right) + (1-\lambda)^2\vec{y}^T A\vec{y}$$

$$= \lambda\vec{x}^T A\vec{x}(\lambda + (1-\lambda)) + (1-\lambda)\vec{y}^T A\vec{y}(\lambda + (1-\lambda)) = \lambda\vec{x}^T A\vec{x} + (1-\lambda)\vec{y}^T A\vec{y} = \lambda f(\vec{x}) + (1-\lambda) f(\vec{y})$$

Thus f is not convex.

Thm 1

Let $f_i : \mathbb{R}^n \to \mathbb{R}$ be a finite collection of convex functions. Then

$$f(\vec{x}) = \max_i f_i(\vec{x})$$

is also convex.

Proof

$$f(\lambda\vec{x} + (1-\lambda)\vec{y}) = \max_i f_i(\lambda\vec{x} + (1-\lambda)\vec{y}) \leq \lambda \max_i f_i(\vec{x}) + (1-\lambda)\max_i f_i(\vec{y}) = \lambda f(\vec{x}) + (1-\lambda) f(\vec{y})$$

Concave Functions

We have a similar class of functions called, **Concave Functions**. A function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be **concave** if each $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $\lambda \in [0,1]$ obey the following inequality:

$$f(\lambda\vec{x} + (1-\lambda)\vec{y}) \geq \lambda f(\vec{x}) + (1-\lambda) f(\vec{y})$$

It turns out that concave functions have many of the same properties as convex functions (or mirrored properties). Affine functions are also concave (proof left for reader). A is negative semi-definite ($f(\vec{x}) = \vec{x}^T A\vec{x} \leq 0 \ \forall x \in \mathbb{R}^n$) if and only if $f(\vec{x})$ is concave. We will not work with concave functions much, but note that many of the convex optimization strategies can be applied for concave objective functions as well.